Cristhian Victor Rojas Marquez

Software Engineer | Full-Stack & AI-Assisted Development

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Professional Summary

Systems Engineering Honors graduate and ICPC national finalist with hands-on experience building full-stack web applications, RESTful APIs, and automated code analysis tools. Strong foundation in Object-Oriented Programming (OOP), data structures, unit/dynamic testing, and modern frameworks (**React, Next.js, NestJS, FastAPI**). Adept at leveraging **AI-assisted development workflows** and LLM SDKs to ship clean, tested, and maintainable software at high velocity within agile teams.

Technical Skills

Programming Languages: TypeScript, JavaScript (ES6+), Python, C++, Java, SQL
Frameworks & Databases: React, Next.js, NestJS, FastAPI, Node.js, Tailwind CSS, PostgreSQL, SQLite, SQLAlchemy, TypeORM
AI & Developer Tools: Cursor, GitHub Copilot, Claude Code, Google GenAI SDK (Gemini API), Git, Docker, Linux
Engineering Practices: RESTful APIs, OOP, Unit & Dynamic Testing, Debugging, Data Structures & Algorithms, CI/CD, Scrum
Languages: Spanish (Native), English (Advanced – Professional Working Proficiency)

Technical Experience

Research Assistant – Software Testing & Dynamic Analysis **Aug 2025 – Dec 2025**
Pontificia Universidad Católica de Chile *Remote*

- Engineered a **Python**-based dynamic code analysis tool to evaluate software reliability and measure the effectiveness of automated test suites.
- Automated runtime execution tracing and metric extraction to detect silent test failures where assertions did not match actual program behavior, streamlining debugging workflows.

Full-Stack Software Developer (Freelance) **Jul 2025 – Nov 2025**
Enterprise Administration System *Cochabamba, Bolivia*

- Collaborated within an agile development team to design and build a full-stack enterprise administration platform using **React**, **NestJS**, and **TypeScript**.
- Architected and consumed modular **REST APIs** applying strict **OOP** design principles, implementing end-to-end business workflows, validation layers, and relational data persistence.

Teaching Assistant & Competitive Programming Mentor **Feb 2023 – Dec 2025**
Universidad Católica Boliviana "San Pablo" *Cochabamba, Bolivia*

- Instructed undergraduate laboratory sessions and conducted code reviews for **Data Structures**, **Programming I (OOP)**, and **Introduction to Programming** across 6 semesters.
- Mentored 20+ engineering students in algorithmic problem-solving, complexity analysis, and systematic debugging for competitive programming.
- Co-organized 4 internal university programming contests, designing algorithmic problem statements, edge-case test suites, and automated grading setups.

Technical Projects

Librerías Bolivia – Book Price Aggregator & API | *FastAPI, Next.js, TypeScript, PostgreSQL* **GitHub Repository**

- Built a full-stack web platform and **RESTful API** consolidating real-time book pricing and availability across **14 Bolivian bookstores and publishers**.
- Developed automated data ingestion adapters consuming **Shopify** and **WooCommerce REST APIs**, paired with a 4-level cascade deduplication and publisher-normalization engine.
- Implemented a responsive **Next.js (React, TypeScript, Tailwind CSS)** frontend and fuzzy-search queries in **PostgreSQL** (`pg_trgm`, `unaccent`), backed by automated unit tests for API endpoints and data normalizers.

Escriba – AI Spanish Linguistic Analyzer | *Python, FastAPI, Gemini 2.5 SDK, Tailwind CSS* **GitHub Repository**

- Developed an AI-native web application integrating the **Google GenAI SDK (Gemini 2.5 Flash)** via a **FastAPI** REST backend to analyze texts and classify errors across spelling, grammar, punctuation, semantics, and style.
- Engineered structured JSON output schemas and research-grounded system prompts (*arXiv:2511.01615*) with IP rate limiting and an interactive inline correction UI.

Education

Universidad Católica Boliviana "San Pablo" **Feb 2021 – Feb 2026**
Bachelor of Science in Systems Engineering – Graduated with Honors (Graduation by Excellence) *Cochabamba, Bolivia*

Achievements & Leadership

- **4th Place Nationwide – ICPC South America South Finals (Nov 2025):** Ranked among the top university competitive programming teams in Bolivia solving complex algorithmic problems under strict time constraints.
- **President – Systems Engineering Scientific Society, SCEIS UCB (Feb 2024 – Dec 2025):** Led student research initiatives, technical workshops, and academic programming events for the Systems Engineering department.
- **Competitive Programming Honors: Silver Medal** at "SanSi Cup 2023" Programming Contest (Dec 2023) & **7th Place Nationwide** at the 2023 ICPC Bolivia Preliminary Contest (Sep 2023).